Guided Inquiry: Reactor Sizing via Levenspiel Plots

Target Audience: Chemical Engineering Juniors (Kinetics & Reactor Design)
Tool: Interactive Levenspiel Plot Explorer

Part 0: The Setup & Theory Recap

Objective: Relate the visual areas on a Levenspiel plot to the integral and algebraic design equations for Ideal Reactors.

Recall the design equations in terms of conversion (X):

1. **CSTR:** $V = \left(\frac{F_{A0} \cdot X}{-r_A}\right)_{\text{exit}}$
2. **PFR:** $V = F_{A0} \int_0^X \frac{dX}{-r_A}$

Open the Interactive Tool:
Look at the Y-axis label. It plots $\frac{F_{A0}}{-r_A}$.

- **Concept Check:** If the Y-value is *high*, is the reaction rate fast or slow?
  - *Answer: High Y-value means a large reciprocal, so the rate ($-r_A$) is SLOW.*

Part 1: Standard Kinetics (Normal Decay)

Action: Select **"Normal Decay"** (Scenario 1) in the tool.

This curve represents typical n^{th} order kinetics (e.g., $A \rightarrow B$) where reaction rate decreases as concentration decreases.

Data Collection:

1. Set the slider to **X = 0.40**.
   - Record V_{CSTR}: ____________
   - Record V_{PFR}: ____________
2. Set the slider to **X = 0.80**.
   - Record V_{CSTR}: ____________
   - Record V_{PFR}: ____________

Analysis:

3. Observe the shape of the curve. Does the value of $\frac{F_{A0}}{-r_A}$ increase or decrease as conversion increases?
4. **Visual Proof:** Why is the CSTR volume (red rectangle) always larger than the PFR volume (blue shaded area) for this scenario?
   * *Hint: Consider where the "missing" triangle is located.*
5. **Chemical Insight:** In a CSTR targeting $X = 0.8$, what is the reaction rate inside the *entire* tank? How does this compare to the average rate inside a PFR operating from $X = 0$ to $X = 0.8$?

Part 2: The Autocatalytic Anomaly

Action: Select **"Autocatalytic"** (Scenario 2) in the tool.

This curve represents a reaction that is self-accelerating (e.g., $A + B \to 2B$) or highly exothermic reactions with thermal effects.

Observation:

6. Move the slider from $X = 0$ to $X = 0.9$. Describe the shape of the curve.
   * At what conversion is the value of $\frac{F_{A0}}{-r_A}$ the **lowest** (i.e., rate is fastest)? $X \approx$ ____________

Data Collection:

7. Set the slider to **X = 0.20** (Before the minimum).
   - Record V_{CSTR}: ____________
   - Record V_{PFR}: ____________
   - **Crucial Question:** Which reactor is smaller? Why?
8. Set the slider to **X = 0.90**.
   - Record V_{CSTR}: ____________
   - Record V_{PFR}: ____________

Analysis:

9. **The CSTR Advantage:** Explain why the CSTR volume is smaller than the PFR volume at low conversions ($X < 0.3$) for this specific kinetics.
   * *Hint: Compare the rate at the exit ($X = 0.2$) vs. the rate at the entrance ($X = 0$).*

Part 3: Engineering Optimization (Reactor Series)

Scenario: You are the lead engineer designing a system to achieve **90% conversion (X=0.9)** for the Autocatalytic reaction (Scenario 2).

You have access to both CSTRs and PFRs and can arrange them in series.

10. **Single Reactor:** Look at your data from Q8. If you use a single reactor, which type saves the most capital cost (Volume)?
11. **Series Configuration (The Challenge):**
    - Look at the curve minimum (approx $X = 0.3$).
    - Imagine using a CSTR to get from $X = 0$ to $X = 0.3$. Draw the rectangle mentally (or calculate $0.3 \times Y_{\text{min}}$).
    - Imagine using a PFR to get from $X = 0.3$ to $X = 0.9$.
    - **Hypothesis:** Would this combination (CSTR then PFR) be smaller than a single PFR or single CSTR?

Conclusion:
Based on the Levenspiel plots, propose a general rule for minimizing total reactor volume:

- Use a **CSTR** when the curve slope is: (Positive / Negative)
- Use a **PFR** when the curve slope is: (Positive / Negative)